13. Wednesday, July 17. The Quotient Topology (§22)

Last Time: 1) Quotient under equivalence relations $X/\sim = \{[x] \mid x \in X\}$

"Paste points that are related" $[x] = \{y \in X \mid y \sim x\}$

2) Have surjective quotient map $\pi: X \to X/\sim$

3) Functions $X/\sim \xrightarrow{\bar{f}} Y$ come from functions $X \xrightarrow{f} Y$ constant on eq. classes and have $\bar{f}(X/\sim) = f(X)$

4) If $f(x) = f(y) \iff x \sim y$, Then $\bar{f}$ is injective

Today: Assume X is Top. space. Define topology on $X/\sim$

Definition / Theorem 1: Let (X, T) be a topological space. Let $\sim$ eq. relation on X and let $\pi: X \to X/\sim$ be the quotient map.

Then the collection

$$T_\sim = \{ V \subseteq X/\sim \mid \pi^{-1}(V) \text{ is open in } X \}$$

defines a topology on $X/\sim$, called the quotient topology.

Proof: Need to verify the 3 topology axioms.

1) $\phi \in T_\sim$ because $\pi^{-1}(\phi) = \phi \in T$. Also $X/\sim \in T_\sim$ because $\pi^{-1}(X/\sim) = X \in T$.

2) If $\{V_i \mid i \in \Lambda\}$ are all in $T_\sim$, since each $\pi^{-1}(V_i) \in T$ and T is a topology we have

$$\pi^{-1}\left(\bigcup_{i\in\Lambda} V_i\right) = \bigcup_{i\in\Lambda} \underbrace{\pi^{-1}(V_i)}_{\in T} \in T. \quad \text{Hence } \bigcup_{i\in\Lambda} V_i \in T_\sim$$

3) If $V_1, V_2 \in T_\sim$ then $\pi^{-1}(V_1), \pi^{-1}(V_2) \in T$. Hence $\pi^{-1}(V_1) \cap \pi^{-1}(V_2) \in T$.

Therefore $\pi^{-1}(V_1 \cap V_2) = \pi^{-1}(V_1) \cap \pi^{-1}(V_2) \in T$. Hence $V_1 \cap V_2 \in T_\sim$ $\square$

Theorem 2: Let $\sim$ be an equivalence relation on a Top space (X, T). Then the quotient $T_\sim$ is the finest topology such that the quotient map $\pi: X \to X/\sim$ is continuous.

Proof: Note that $\pi: (X, T) \to (X/\sim, T_\sim)$ is continuous. In fact, if $V \subseteq X/\sim$ is in $T_\sim$, then by definition of $T_\sim$ we have $\pi^{-1}(V) \in T$. Now if T' is any topology on $X/\sim$ such that π is continuous, then given any $V \subseteq X/\sim$ with $V \in T'$, we have $\pi^{-1}(V) \in T$. Hence $V \in T_\sim$ by definition of $T_\sim$. Hence $T' \subseteq T_\sim$, as desired $\square$.

Theorem 3: Let $f: X \to Y$ is a map between Top spaces. Assume that f descends to $\bar{f}: X/\sim \to Y$. Then the following hold:

1) f is continuous if and only if $\bar{f}$ is continuous.
2) If f is open, then $\bar{f}$ is open.

Proof: 1) ($\Leftarrow$) Assume $\bar{f}$ continuous. Since $f = \bar{f} \circ \pi$ and π is continuous, it follows that f is continuous.

($\Rightarrow$) Assume f is continuous. WTS $\bar{f}: X/_\sim \to Y$ is continuous. Let $U \subseteq Y$ open.

Want to show $\bar{f}^{-1}(U) \subseteq X/_\sim$ continuous. But by def^n of quotient topology, that means $\pi^{-1}\left(\bar{f}^{-1}(U)\right) \subseteq X$ is open. Since $f = \bar{f} \circ \pi$, we have

$$\pi^{-1}\left(\bar{f}^{-1}(U)\right) = \left(\bar{f} \circ \pi\right)^{-1}(U) = f^{-1}(U)$$

And since f is continuous, we know $f^{-1}(U)$ is open in X.

Hence $\bar{f}^{-1}(U) \subseteq X/_\sim$ is open for any open $U \subseteq Y$.

Hence $\bar{f}$ is continuous.

2) HW!

$\square$

Example: Consider $X = \mathbb{R}^2$ with the standard topology

Define a relation on X by $(x_1, x_2) \sim (y_1, y_2) \iff x_1^2 + x_2^2 = y_1^2 + y_2^2$ (ie: same distance to origin)

Then the set of equivalence classes is

$$X/_\sim = \{\text{Circles centered at the origin}\} \cup \{\{0\}\}$$

Claim: There is a homeomorphism $X/_\sim \longrightarrow [0, \infty)$

Define first a map $f: X \longrightarrow [0, \infty)$ by $f(x_1, x_2) = \sqrt{x_1^2 + x_2^2} = \|x\|$ Notice that

1) If $(x_1, x_2) \sim (y_1, y_2) \Rightarrow x_1^2 + x_2^2 = y_1^2 + y_2^2 \Rightarrow f(x_1, x_2) = f(y_1, y_2)$

Hence f descends to a map $\bar{f}: X/_\sim \longrightarrow [0, \infty)$ given by $\bar{f}([(x_1, x_2)]) = f(x_1, x_2)$.

2) $\bar{f}$ is injective: in fact if $f(x_1, x_2) = f(y_1, y_2)$, then $(x_1, x_2) \sim (y_1, y_2)$

3) $\bar{f}$ is surjective, because f is surjective

4) $\bar{f}$ is continuous, because f is continuous (know this from analysis)

5) $\bar{f}$ is open. By the previous Thm, we just need to prove that f is open.

And for this, just need to show that $f(B(x,\varepsilon)) \subseteq [0,\infty)$ is open $\forall x \in X$ $\forall \varepsilon > 0$.

But it is easy to prove that

$$f(B(x,r)) = \begin{cases} (\|x\| - r,\ \|x\| + r) & \text{if } \|x\| \geq r \\ [0,\ \|x\| + r) & \text{if } \|x\| < r \end{cases}$$

Which, in any case, is open in $[0,\infty)$.

Therefore $\bar{f} : X/_{\sim} \longrightarrow [0,\infty)$ where $\bar{f}([x]) = \|x\|$ is a homeomorphism. $\square$